Image Processing & Computer Vision

A 3-day PyTorch course

Markus Hinsche · datascienceretreat.com · 2026

Agenda

Over three days we'll cover:

- **Day 1** — images as tensors, convolutions, datasets & augmentations
- **Day 2** — building and training CNNs, transfer learning, Grad-CAM
- **Day 3** — attention, building a ViT from scratch, CNN vs ViT

Hands-on throughout. Every concept is paired with a notebook you run yourself.

Topics we can dig into on request:
mixed-precision training, multi-GPU, model ensembling, hyperparameter tuning,
data-centric vs model-centric iteration, deployment.

Download slides and code

Clone the repo to follow along:

```
$ git clone https://github.com/markus-hinsche/img-proc-and-comp-vision.git
$ cd img-proc-and-comp-vision
$ uv sync
$ uv run jupyter lab
```

Notebooks live in `notebooks/`, slides in `slides/`.

whoami

- **Markus Hinsche** — co-founder & CTO of Thea Care
  - B2B SaaS: AI skin analysis for cosmetic and pharma brands
  - Computer vision in production (segmentation + classification)
- Previously: ML freelancer — Welthungerhilfe, Charité radiology (both CV)
- Strengths:
  - Python (since 2009), Deep Learning (since 2018)
  - Productionizing models, MLOps, mentoring
- markushinsche.de · github

Why computer vision?

"Vision is the killer app of deep learning."

Image classification was the first task where deep nets clearly beat classical methods (AlexNet, 2012). The recipes you learn this week — convolutions, transfer learning, attention — are the foundation for everything from medical imaging to robotics to generative models.

What you'll be able to do by Friday

- Load any image dataset into PyTorch and apply augmentations
- Build a CNN from scratch and train it on GPU
- Fine-tune a pretrained model (ResNet, EfficientNet, ViT) on your own data
- Debug training failures (loss not decreasing, overfitting, wrong shapes)
- Explain *why* a model predicts what it predicts (Grad-CAM)
- Understand the bridge from CNNs to Vision Transformers

Ground rules

- **Ask questions immediately.** If you're lost on slide 3, you'll be lost on slide 30.
- **Run the code.** Reading notebooks is not the same as running them.
- **Pair up.** Two brains debug faster than one.
- **It's OK if your model doesn't train.** Half the skill is figuring out why.

Let's go.

Open `notebooks/01_images_as_tensors.ipynb`